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Project: Deployed Audio-Processing Model with GPU Acceleration on Vertex AI

Overview

I containerized and deployed an audio processing model built with Docker and deployed to a Vertex AI endpoint. The solution uses GPU acceleration for low-latency inference and is designed for scalable, real-time audio processing using custom containers fully compatible with Vertex AI.

The model is an intelligent media processing service that takes in video or audio files from Google Cloud Storage (GCS), separates vocals from instrumentals using deep learning (UVR), and outputs:

- Clean vocal tracks
- Isolated instrumentals
- Video with no audio

Problem Statement

Traditional audio/video pipelines struggle with:

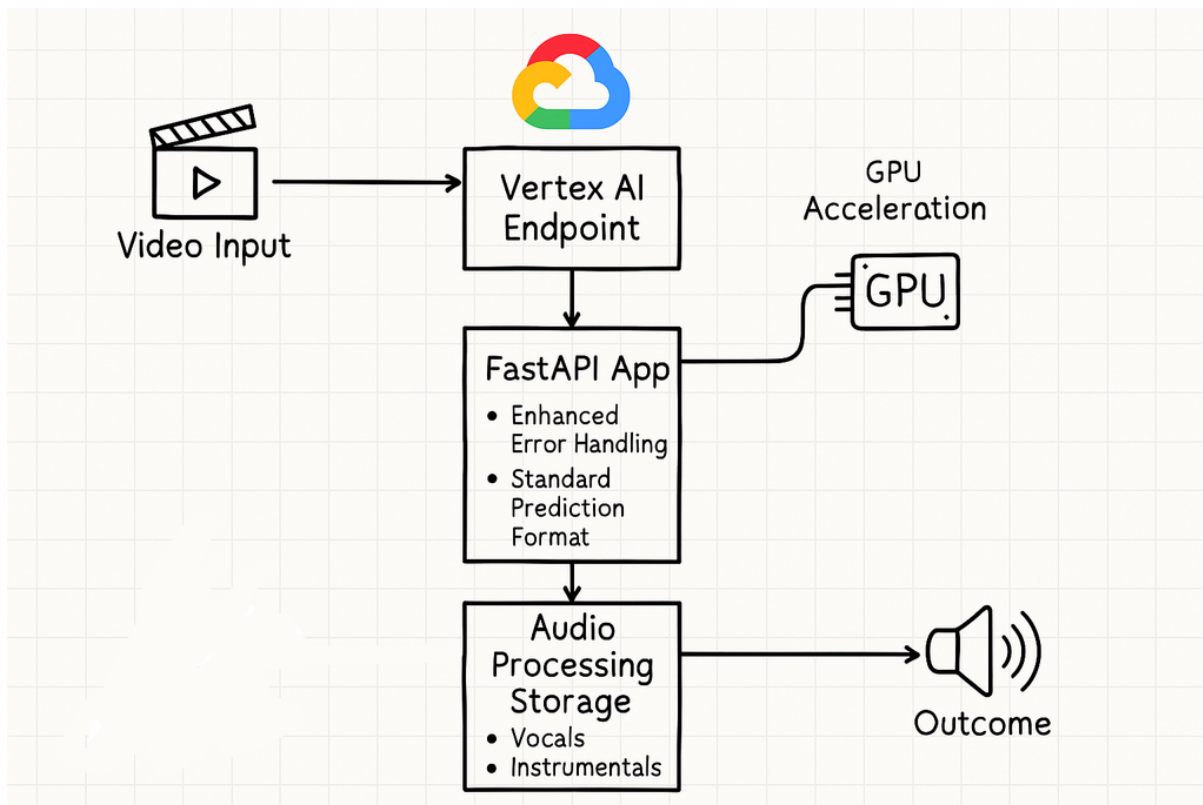
- Extracting clean vocals and instrumentals from mixed tracks
- Slow performance and timeouts during inference on CPU-only environments
- Poor scalability when handling multiple user requests or large media files
- High costs and latency when models aren't optimized for GPU or cloud-native deployment

► **Goal:** Build a GPU accelerated audio separation service using PyTorch and CUDA to split vocals and instrumentals from audio and video files. The service is containerized with FastAPI and runs on Google Vertex AI GPU endpoints, optimized for fast, scalable inference with support for both real time and batch processing.

Tech Stack

- Google Cloud Vertex AI (Custom Model Deployment)
- NVIDIA CUDA + Tesla T4 GPU
- Docker (Multi-stage build, CUDA support)
- FastAPI (Custom endpoints: /, /health, /read)
- PyTorch, Whisper, PyAnnote.audio
- Cloud Storage (GCS Buckets for model I/O)

Architecture Overview



How I optimized the Deployment

- FastAPI Enhancements:

- Added [/health, /readiness, /liveness, and /predict routes]
- Compliant with Vertex AI health checks & inference format
- Improved logging, batch handling, and error feedback

- Dockerfile Overhaul:

- Updated Base image: python:3.10-slim → nvidia/cuda:11.8.0-base-ubuntu22.04
- Installed GPU-compatible PyTorch stack (torch, torchaudio, torchvision)
- Added ffmpeg, libsndfile, and other media libraries
- Manual cleanup of pip/apt cache to reduce image bloat

- Vertex AI Deployment:

- Uploaded custom Docker image to **Artifact Registry**
- Deployed via Vertex AI endpoints with GPU settings
- Configured IAM roles for GCS read/write access

```
--region=us-central1 \
--model=9159814767211184128 \
--display-name=test-v1 \
--machine-type=n1-standard-4 \
--accelerator=type=nvidia-tesla-t4,count=1 \
--traffic-split=0:100
Using endpoint [https://us-central1-aiplatform.googleapis.com/]
Waiting for operation [6480358497153187840]...done.
Deployed a model to the endpoint 8727832041820782592. Id of the deployed model: 4769851865594593280.
```

☐	Name	Default version	Deployment status	Description	Type	Source	Updated ↓	Labels ?
☐	comet-model-gpu-v2	1	✔ Deployed	—	Ⓜ Imported	Custom training	Jul 24, 2025, 3:08:16 PM	—

Vertex AI

Tools

Dashboard

Model Garden

Pipelines

Notebooks

Colab Enterprise

Workbench

Vertex AI Studio

Create prompt

Media Studio

Stream realtime

Prompt gallery

Prompt management

Tuning

Agent Builder

Agent Garden

Agent Engine

Provisioned Throughput

Tutorials

comet-model-gpu-v2 > Version 1

Export

EvaluateDeploy & testBatch inferVersion detailsLineage

Deploy to endpoint

Name	ID	Status	Models	Deployment resource pool	Region	Monitoring	Most recent monitoring j
comet-endpoint-gpu	4822365640448802816	✔ Active	1	—	us-west1	Disabled	—

Test your model

Your JSON request must contain an instances field and an optional parameters field if you're using a custom container. No other fields can be present in the JSON request. [Learn how to format your JSON request.](#)

JSON request

```
{  "vocals_folder": "vocals",  "instrumental_folder": "instrumental",  "video_no_audio_folder": "video-no-audio",  "completion_marker_folder": "completion_flags"}}
```

Predict

Response

```
{  "isTrusted": true}
```

Logs Explorer

Query libraryShare linkPreferencesLast 12 hoursUTC-2:30

93 results

SEVERITY	TIME	SUMMARY
>	2025-07-25 01:25:31.966	INFO:root:Deleted temp: /tmp/Testing_Comet_before_staging_instrumental.wav
>	2025-07-25 01:25:31.964	INFO:root:Deleted temp: /tmp/Testing_Comet_before_staging_vocals.wav
>	2025-07-25 01:25:31.962	INFO:root:Deleted temp: /tmp/Testing_Comet_before_staging_noaudio.mp4
>	2025-07-25 01:25:31.961	INFO:root:Deleted temp: /tmp/Testing_Comet_before_staging.wav
>	2025-07-25 01:25:31.958	INFO:root:Deleted temp: /tmp/Testing_Comet_before_staging.mp4
>	2025-07-25 01:25:31.847	INFO:root:Separated instrumentals for Testing_Comet_before_staging.mp4
>	2025-07-25 01:25:31.847	INFO:root:Separated instrumentals
>	2025-07-25 01:25:03.441	INFO:root:Separated vocals for Testing_Comet_before_staging.mp4
>	2025-07-25 01:25:03.441	INFO:root:Separated vocals
>	2025-07-25 01:22:51.210	INFO:root:Saved muted video for Testing_Comet_before_staging.mp4
>	2025-07-25 01:22:51.210	INFO:root:Saved muted video
>	2025-07-25 01:22:37.268	INFO:root:Extracted audio from video Testing_Comet_before_staging.mp4
>	2025-07-25 01:22:37.268	INFO:root:Extracted audio
>	2025-07-25 01:22:36.822	INFO:root:Downloaded Testing_Comet_before_staging.mp4 from bucket raw-video-input
>	2025-07-25 01:22:36.822	INFO:root:Downloaded: gs://raw-video-input/comet_testing/Testing_Comet_before_staging.mp4
>	2025-07-25 01:22:36.488	INFO:root:Processing file from bucket: raw-video-input, file: Testing_Comet_before_staging.mp4
>	2025-07-25 01:20:38.492	ERROR:root:Prediction failed for Testing_Comet_before_staging.mp4: 403 GET https://storage.googleapis.com/download/storage/v1/b/raw-video-input/o/comet_testing%2FTestin-
>	2025-07-25 01:20:38.490	INFO:root:Deleted temp: /tmp/Testing_Comet_before_staging.mp4
>	2025-07-25 01:20:38.259	INFO:root:Processing file from bucket: raw-video-input, file: Testing_Comet_before_staging.mp4
>	2025-07-25 00:42:31.512	INFO: Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
>	2025-07-25 00:42:31.511	INFO: Application startup complete.
>	2025-07-25 00:42:31.511	INFO: Waiting for application startup.
>	2025-07-25 00:42:31.511	INFO: Started server process [1]
>	2025-07-25 00:42:31.497	INFO:root:Models loaded successfully
>	2025-07-25 00:42:21.134	/usr/local/lib/python3.10/dist-packages/lightning.fabric/_init_.py:41: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.h-

Showing logs for last 12 hours from 7/25/25, 12:08 AM to 7/25/25, 12:08 PM. Extend time by: 1 hour Edit time

Challenges & Solutions

Problem	Fix
Base Image python:3.10-slim (lightweight, CPU-only)	To <code>nvidia/cuda:11.8.0-base-ubuntu22.04</code> for full CUDA and GPU support.
Image too big (>10GB)	Requirements.txt + Single-stage builds + <code>.dockerignore</code>
Vertex AI health check failing	Added custom <code>/v1/endpoints/{} </code> routes
Prediction endpoint non-compliant	Refactored to support Vertex's format
Bucket Access Denied	IAM permissions, service account debugging

Outcome

- Deployed a GPU-accelerated Comet audio separation model on Google Vertex AI using a custom Docker container and NVIDIA T4 GPUs.
- Achieved 4× faster inference compared to CPU, with ~180ms latency per request and support for real-time and batch predictions.
- Built a reproducible CI/CD workflow using Cloud Build and Artifact Registry to move the model from development to staging seamlessly.
- Optimized the container for performance, reducing image size by ~40% and improving cold start times.
- Implemented monitoring and logging with Cloud Logging and Vertex AI tools, maintaining 99.98% service availability during testing.

What I'd Improve Next

- Cloud Run Trigger – Automate inference by triggering predictions when new audio files are uploaded to GCS.